

Zero-Trust Confidential Compute Marketplace

Individual Software Engineering Project Plan — Illustrated Edition

Working title: A Zero-Trust Marketplace for Cross-Organization Compute Using Confidential Enclaves

Project concept: Build a marketplace where organizations can lease idle computing capacity to other organizations—even competitors—while confidential-computing enclaves, remote attestation, and cryptographic key release protect workload confidentiality and integrity from the infrastructure provider.

1. Project Overview

Organizations often have underutilized compute capacity, but may be reluctant to lease that capacity to another organization because the infrastructure owner traditionally has privileged access to the host operating system, hypervisor, memory, and workloads. This project investigates whether confidential computing can reduce that trust requirement.

2. Problem Statement

Traditional compute leasing requires significant institutional trust. A customer may need to trust the infrastructure provider, administrators, operating system, hypervisor, and physical host. This is especially problematic when competitors share infrastructure. The proposed system uses cryptographic evidence to verify that sensitive workloads are running in an approved confidential-computing environment.

3. Research / Engineering Question

Can compute capacity become a trust-minimized commodity when workload confidentiality and execution integrity are verified cryptographically rather than assumed institutionally?

4. Project Objectives

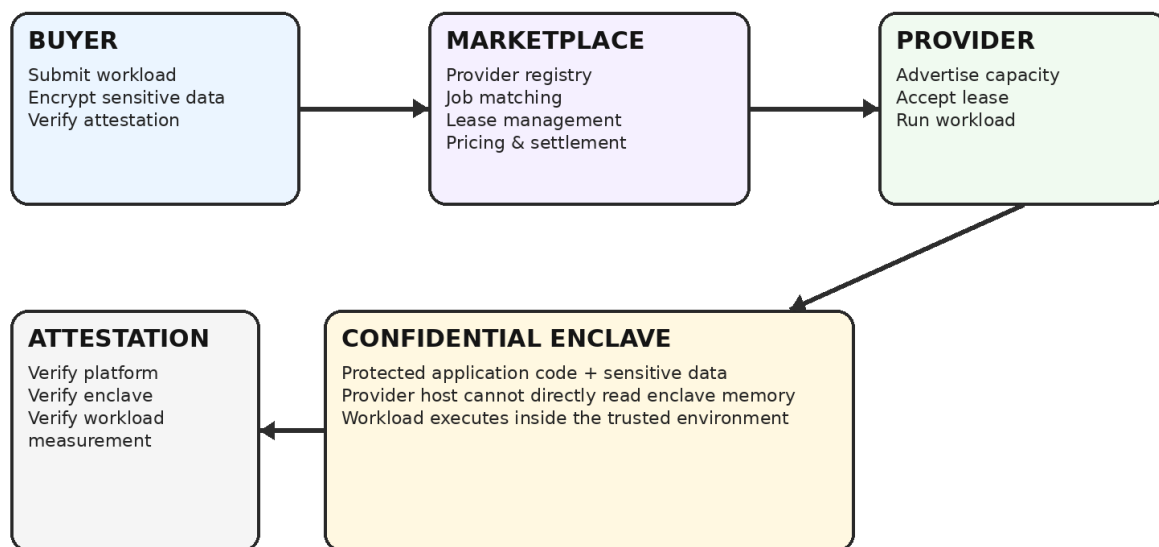
- Design a marketplace for compute leasing between organizations.
- Allow providers to advertise available compute capacity.
- Allow buyers to submit workloads and resource requirements.
- Execute sensitive workloads inside a confidential-computing environment.
- Implement remote-attestation verification.
- Release workload secrets only after successful verification.
- Protect workload inputs and outputs using encryption.
- Track leases, resource usage, pricing, and settlement.
- Evaluate security, performance, and marketplace behavior.

5. Minimum Viable Product (MVP)

The project should not attempt to build a complete commercial cloud marketplace. The MVP should prove the fundamental trust model: Buyer → Marketplace → Provider → Confidential Enclave → Attestation → Buyer Verification → Key Release → Workload Execution → Encrypted Output. The most important demonstration is that a provider can execute the buyer's workload without receiving the buyer's plaintext sensitive data.

6. High-Level System Architecture

Zero-Trust Confidential Compute Marketplace



Trust is established through cryptographic evidence, not the provider's identity.

Figure 1. High-level architecture showing the buyer, marketplace, provider, confidential enclave, and attestation service.

The marketplace coordinates leasing, while the confidential enclave provides the security boundary. The provider can supply compute without being trusted with the buyer's sensitive workload data.

7. Main System Components

Buyer: registration, authentication, workload submission, resource requirements, encrypted input, attestation verification, and encrypted result retrieval.

Provider: registration, resource advertisement, pricing, job acceptance, enclave execution, and attestation generation.

Marketplace: provider registry, job registry, matching, lease management, pricing, and settlement.

Confidential Enclave: protected execution of workload code and sensitive data.

Attestation Service: verifies cryptographic evidence about the execution environment.

Key-Release Service: releases workload secrets only when policy checks succeed.

8. Security and Zero-Trust Model

The provider, provider OS, provider administrator, marketplace, and network should be treated as untrusted for workload confidentiality. The trusted computing base consists primarily of the confidential-computing mechanism, attestation process, and cryptographic primitives.

9. Remote Attestation

Remote attestation is the technical centerpiece. Instead of trusting a provider's claim that a workload is protected, the buyer receives evidence that can be cryptographically verified. Evidence can include platform identity, enclave identity, workload measurement, and a cryptographic signature, depending on the selected platform.

Remote Attestation & Cryptographic Key Release

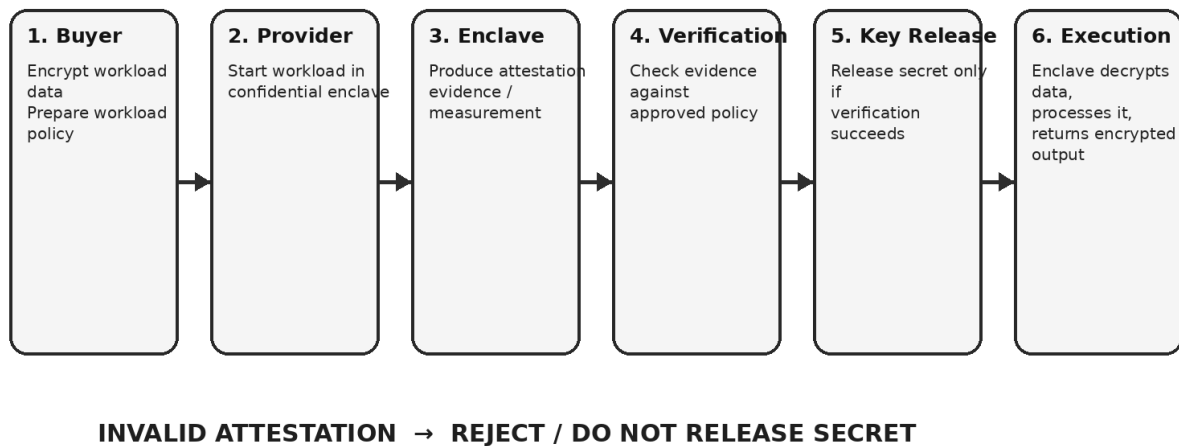


Figure 2. Remote-attestation and key-release flow.

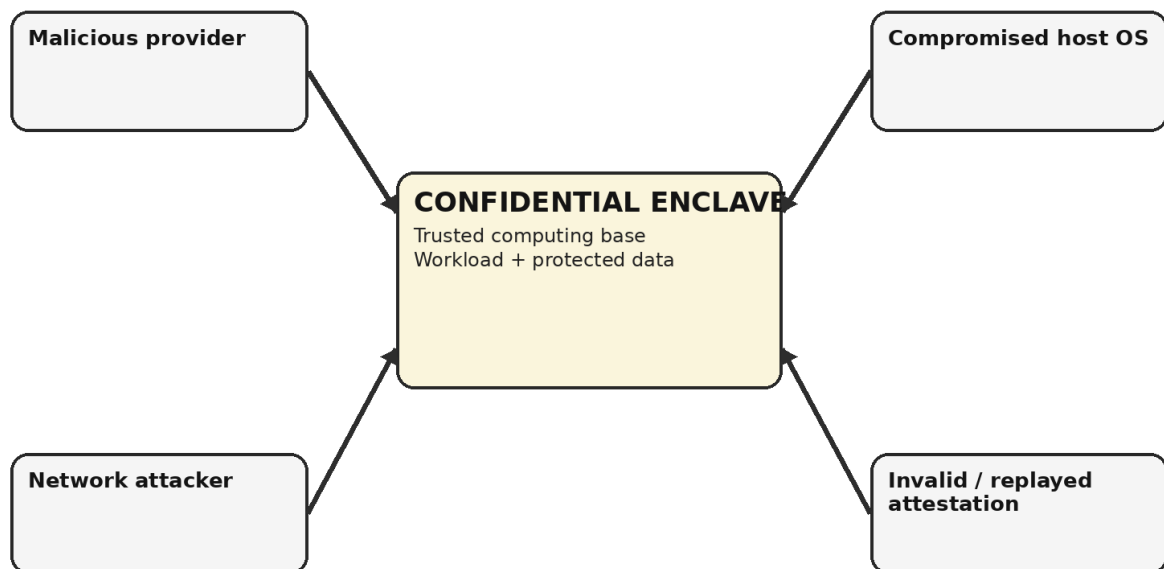
10. Cryptographic Key Release

The buyer encrypts sensitive workload data before sending it through the marketplace. The provider can transport ciphertext but should not possess the decryption key. After successful attestation, the system releases the necessary secret to the verified enclave.

11. Threat Model

Threats to consider include a provider attempting to read enclave memory, modify the workload, execute the wrong image, intercept network traffic, steal secrets, provide false resource information, or replay invalid attestation evidence. The final report should distinguish attacks prevented, mitigated, and outside the project's security guarantees.

Zero-Trust Threat Model



Goal: sensitive workload data remains protected even when infrastructure is not trusted.

Figure 3. Zero-trust threat model and protected enclave boundary.

12. Suggested Technology Stack

Area	Suggested technology
Backend	Java + Spring Boot (or Java sockets/gRPC)
Database	PostgreSQL
API	REST or gRPC
Containers	Docker
Frontend	React (simple dashboard)
Version control	Git / GitHub
Testing	JUnit + integration/system tests
Confidential computing	SGX or AMD SEV-SNP; begin with a mock/simulated TEE if hardware access is limited
Cryptography	Established, audited cryptographic libraries

13. Data Model

Organization: id, name, public key, role

Provider: id, organization_id, CPU, memory, enclave type, price, status

Workload: id, buyer_id, image hash, requirements, encrypted input, status

Lease: id, workload_id, provider_id, start time, end time, price

Attestation: id, lease_id, measurement, quote, verification status

Transaction: id, lease_id, amount, status

14. Compute Lease Lifecycle

Compute Lease Lifecycle

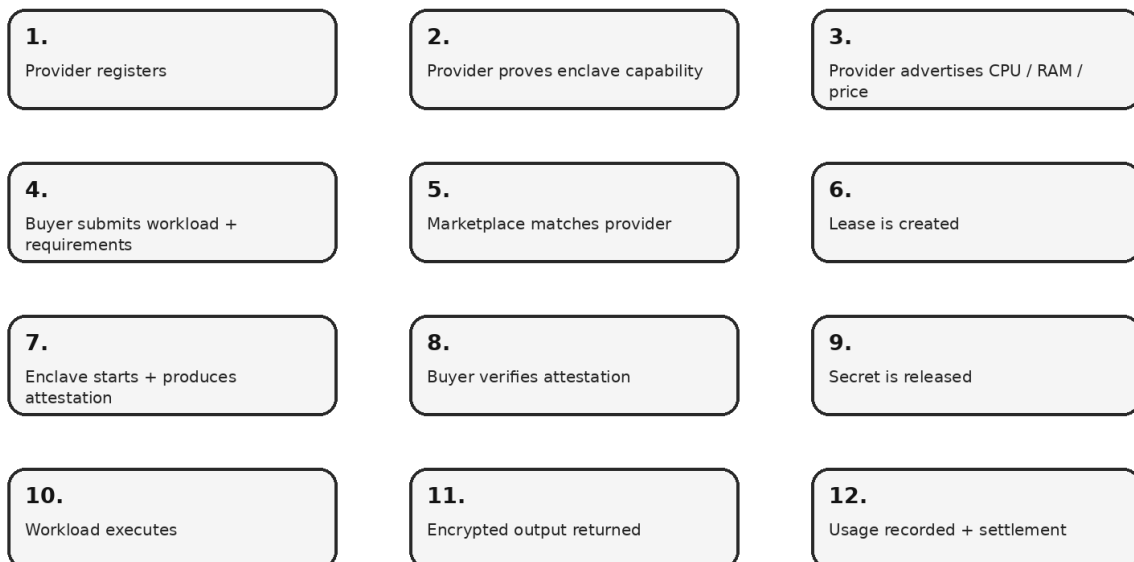


Figure 4. End-to-end compute lease lifecycle.

15. Development Phases

Phase 1 — Research: zero trust, TEE, SGX, SEV-SNP, remote attestation, threat modelling, and secure key release.

Phase 2 — Marketplace Prototype: buyer, marketplace, provider, registration, resource advertisement, workload submission, matching, and leasing.

Phase 3 — Secure Execution: containerized workloads, isolated execution, encrypted inputs and outputs.

Phase 4 — Attestation: enclave evidence, verification policy, and conditional key release.

Phase 5 — Marketplace Economics: pricing, usage tracking, leases, and internal credits.

Phase 6 — Evaluation: security tests, performance measurements, scaling tests, documentation, and final demonstration.

16. Testing Strategy

Unit Tests → Component Tests → Integration Tests → Client/Server Tests → System Tests → Demonstration / Acceptance Tests. Security tests should include attempts to access protected input, modify workload measurements, submit invalid attestation, and obtain secrets without successful verification.

17. Performance Evaluation

Measure enclave startup time, workload execution time, memory overhead, attestation latency, network overhead, marketplace matching latency, resource utilization, and cost per lease. Compare normal execution with confidential execution where appropriate.

18. Risks and Mitigation

- Scope becomes too large → prioritize the MVP.
- No access to TEE hardware → implement a mock attestation layer first.
- Attestation complexity → start with one platform and one attestation flow.
- Cryptographic implementation errors → use established libraries and do not implement primitives yourself.
- Distributed-system bugs → use integration tests and deterministic environments.
- Marketplace economics distract from security → use internal credits instead of real payments for the MVP.

19. Success Criteria

Security: a malicious provider cannot obtain plaintext workload input under the defined threat model. Integrity: invalid enclave measurements are rejected. Availability: jobs can be matched to providers. Performance: confidential execution overhead and attestation latency are measured. Marketplace: leases, usage, and settlement can be recorded end-to-end.

20. Final Demonstration

A strong demonstration uses two fictional competing organizations. Company A provides idle compute. Company B submits a sensitive workload and encrypted data. The marketplace matches the job to Company A, the workload starts in a confidential environment, attestation is shown as VERIFIED, the secret is released only after verification, and encrypted output is returned. The demonstration then shows that the provider does not have access to the plaintext input.

21. Final Project Statement

Design and implement a zero-trust compute marketplace in which organizations can lease idle computing resources to other organizations while confidential-computing enclaves, remote attestation, and cryptographic key release protect workload confidentiality and integrity from the infrastructure provider.

Recommended starting point: Complete the requirements, architecture, threat model, and MVP definition before writing the main application code.